



IT582: Foundations of Machine Learning

Lab-3: Feature Scaling and Regularized Linear Regression

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Submission Instructions

Upload **only one Jupyter Notebook (.ipynb)** file to Moodle.

The notebook should contain all code, outputs, plots, observations and conclusions.

Name the notebook as:

[studentid]_lab3.ipynb

Task 1: Data Preparation and Feature Scaling

Use the **California Housing Dataset** with all available input features and `MedHouseValue` as the target.

Split the dataset into:

- 60% Training data
- 20% Validation data
- 20% Test data

For every input feature x_j , perform standardization using

$$x_j^{(\text{scaled})} = \frac{x_j - \mu_j}{\sigma_j},$$

where μ_j and σ_j are calculated using the **training data only**.

Use the same training-set values of μ_j and σ_j to transform the validation and test sets.

Verify that the scaled training features have approximately zero mean and unit standard deviation.

Task 2: Effect of Feature Scaling on Gradient Descent

Consider the Linear Regression model

$$h_{\boldsymbol{\theta}}(\mathbf{x}) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d.$$

Use the cost function

$$J(\boldsymbol{\theta}) = \frac{1}{2n} \sum_{i=1}^n (h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - y^{(i)})^2.$$

Implement **Batch Gradient Descent** for Linear Regression.

Train the model using:

- Original unscaled features
- Scaled features

Use the same initial parameter values in both cases. Choose suitable learning rates so that Gradient Descent converges.

Plot the training loss as a function of iteration for both cases.

Compare:

- Convergence behaviour
- Number of iterations required for convergence
- Execution time

Write a short conclusion on these comparisons.

Use the **scaled features** for all remaining tasks.

Task 3: Ridge Regression using Gradient Descent

For Ridge Regression, use the regularized cost function

$$J(\boldsymbol{\theta}) = \frac{1}{2n} \left[\sum_{i=1}^n (h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^d \theta_j^2 \right].$$

Here,

$$\lambda \geq 0$$

is the regularization parameter.

The intercept parameter θ_0 should **not** be regularized. For θ_0 , use the update

$$\theta_0 := \theta_0 - \alpha \frac{1}{n} \sum_{i=1}^n (h_{\theta}(\mathbf{x}^{(i)}) - y^{(i)}) .$$

For $j = 1, 2, \dots, d$,

$$\theta_j := \theta_j - \alpha \left[\frac{1}{n} \sum_{i=1}^n (h_{\theta}(\mathbf{x}^{(i)}) - y^{(i)}) x_j^{(i)} + \frac{\lambda}{n} \theta_j \right] .$$

Implement Ridge Regression using **Batch Gradient Descent**.

First run the implementation using

$$\lambda = 0.$$

Verify that for $\lambda = 0$, the regularized Gradient Descent implementation reduces to ordinary Linear Regression.

Task 4: Significance and Visualization of Regularization

A. Effect of λ on the Regression Model

Using all scaled input features, train the Ridge Regression model for

$$\lambda \in \{0, 0.01, 0.1, 1, 10, 100\} .$$

For each value of λ , record:

- Training MSE
- Validation MSE
- Final model coefficients
- Magnitude of the parameter vector

Calculate the parameter magnitude as

$$\|\boldsymbol{\theta}_{1:d}\|_2 = \sqrt{\theta_1^2 + \theta_2^2 + \dots + \theta_d^2} .$$

Create the following plots:

1. Plot the learned coefficients $\theta_1, \theta_2, \dots, \theta_d$ against λ .
Use a separate curve for each feature coefficient.
2. Plot $\|\boldsymbol{\theta}_{1:d}\|_2$ against λ .
3. Plot Training MSE and Validation MSE against λ .

Write your conclusions on this study

B. Visualization of the Regularized Cost Function

To visualize how regularization changes the cost function, use only the scaled `MedInc` feature and the target `MedHouseValue`.

For this visualization, consider the model

$$h_{\boldsymbol{\theta}}(x) = \theta_0 + \theta_1 x.$$

The corresponding Ridge Regression cost is

$$J(\theta_0, \theta_1) = \frac{1}{2n} \left[\sum_{i=1}^n (\theta_0 + \theta_1 x^{(i)} - y^{(i)})^2 + \lambda \theta_1^2 \right].$$

Evaluate the cost over a suitable grid of values for θ_0 and θ_1 .

Create separate **2D contour plots** of the cost function for:

$$\lambda = 0, \quad \lambda = 1, \quad \lambda = 100.$$

Compare the contour plots and write observations.

Task 5: Ridge Regression using the Normal Equation

For the scaled training data, construct the input matrix $\mathbf{X}$ by adding a column of ones for the intercept term.

Use the regularized Normal Equation

$$\boldsymbol{\theta} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{R})^{-1} \mathbf{X}^T \mathbf{y},$$

where

$$\mathbf{R} = \text{diag}(0, 1, 1, \dots, 1).$$

The first diagonal element of $\mathbf{R}$ is zero so that θ_0 is not regularized.

Train this model using the same hyperparameters as in the implementation with GD.

For both methods, compare:

- Training MSE
- Validation MSE
- Test MSE
- Execution time

Compare the parameter vectors obtained from the two methods using

$$\|\boldsymbol{\theta}_{\text{GD}} - \boldsymbol{\theta}_{\text{NE}}\|_2.$$